Learning Objective 3.9: Calculate null-steering weights, implement pattern nulls on the ADALM-PHASER, and compare manual null steering with adaptive beamforming.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From weights to GUI settings.** A null-steering calculation for a 4-element subarray returns the complex weights

$$w_1 = 0.62 - j0.13, \quad w_2 = 0.88 + j0.05, \quad w_3 = 0.88 - j0.05, \quad w_4 = 0.62 + j0.13$$

The Phaser GUI takes element amplitude as a percentage of the largest weight and element phase as an offset in degrees.

(a) Compute the four magnitudes and convert them to the percentages you would type into Element Gains.

gains = $\boxed{72, 100, 100, 72\ \%}$

$$|w_1| = \sqrt{0.62^2 + 0.13^2} = 0.634$$
$$|w_2| = \sqrt{0.88^2 + 0.05^2} = 0.881$$

Elements 3 and 4 have the same magnitudes as 2 and 1. The largest weight is 0.881, so the percentages are $100|w_n|/0.881$.

$$72\%, \quad 100\%, \quad 100\%, \quad 72\%$$

(b) Compute the four phase offsets in degrees.

phases = $\boxed{-11.9, +3.3, -3.3, +11.9^\circ}$

Each phase is $\angle w_n = \arctan(\mathrm{Im}/\mathrm{Re})$, with both weights in the right half plane so no quadrant correction is needed.

$$\angle w_1 = \arctan(-0.13/0.62) = -11.9^\circ$$
$$\angle w_2 = \arctan(+0.05/0.88) = +3.3^\circ$$

$$-11.9^\circ, \quad +3.3^\circ, \quad -3.3^\circ, \quad +11.9^\circ$$

(c) A student enters the four gain percentages but forgets to enter the phase offsets. Describe the pattern they will measure and say whether a notch appears.

With the phases left at zero the array is a uniform-phase amplitude taper of $72, 100, 100, 72$ percent. That is a mild taper, so the beam broadens slightly and the sidelobes drop by a fraction of a decibel, but the pattern stays symmetric about broadside and no notch appears at the intended angle. The amplitudes alone cannot place a null; the cancellation depends on the relative phases, and setting them all to zero removes it.

(d) All four weights are multiplied by e^{j40° before being entered. How does the measured pattern change, and why?

It does not change. A common phase factor multiplies the entire array sum by a constant of unit magnitude, and the measured quantity is $|F(\theta)|$. Only the phase *differences* between elements affect the pattern, which is why the GUI accepts phase offsets rather than absolute phases.

2. **Reading a null-steered sweep.** A student sweeps the uniform 8-element array, freezes the trace, enters null-steering weights for a null at $+22.5°$, and sweeps again. The two traces read as follows.

	Uniform (frozen)	**Null steered**
Peak level	0.0 dB (reference)	−1.8 dB
Level at $+22.5°$	−12.8 dBc	−21.6 dBc

(a) How much additional rejection did the null steering buy at $+22.5°$?

$\Delta = \boxed{8.8 \text{ dB}}$

$$\Delta = -12.8 \text{ dB} - (-21.6 \text{ dB}) = 8.8 \text{ dB}$$

The response at the interferer angle dropped by 8.8 dB relative to the frozen uniform reference.

(b) Express the notch depth relative to the null-steered array's own peak, which is the number a jammer-to-signal calculation needs.

depth $= \boxed{-19.8 \text{ dBc}}$

Both levels are quoted against the uniform reference peak, so subtract the new peak level from the notch level.

$$-21.6 \text{ dB} - (-1.8 \text{ dB}) = -19.8 \text{ dB}$$

Relative to its own main lobe the array rejects the interferer by 19.8 dB , not 21.6.

(c) A second student reports a notch at −31 dBc on the same bench with the same weights. The sweep's noise floor sits about 23 dB below the uniform peak. Is the report credible? Explain.

No. The sweep cannot display anything below its own noise floor, so no point on the trace can read more than about 23 dB below the uniform reference peak, which is about 21 dB below the null-steered array's own peak once the 1.8 dB main-lobe cost is accounted for. A reported −31 dBc is below the floor and indicates a reading taken from the wrong axis, a different reference, or a mis-set peak marker. The correct statement in that situation is that the notch reached the noise floor and its true depth is not observable with this measurement.

(d) The notch appears centered at $+19.7°$ rather than $+22.5°$. Give the most likely cause.

Phase quantization. The ADAR1000 rounds each commanded phase to the nearest $2.8125°$, so the vector the hardware applies is not the vector that was computed. The rounded vector is still a valid weight vector, and it places its null a step or two away from the requested angle. The sweep grid is the same $2.8125°$ step, which is why the displacement lands on a grid point.

3. **Null depth under coarse quantization.** The lab is repeated on a board whose phase shifters have only $B = 2$ bits.

   (a) Find the phase LSB.

$$\text{LSB} = \frac{360^\circ}{2^B} = \frac{360^\circ}{4} = 90^\circ$$

LSB = $\boxed{90^\circ}$

   (b) Use the rule of thumb that quantization sidelobes sit near $-6B$ dB to estimate the deepest notch this board can hold.

depth ≈ $\boxed{-12 \text{ dB}}$

$$\text{QSLL} \approx -6B = -6(2) = -12 \text{ dB}$$

Quantization scatters roughly 12 dB of energy into the pattern, so a notch cannot be held deeper than about 12 dB below the peak. The 7-bit ADAR1000 scatters far less: its rounded weights still permit a notch near 48 dB down, so the 20 to 22 dB measured in the lab is the sweep's noise floor and not a quantization limit.

   (c) Round the four phases from Question 1(b) to the nearest multiple of the 90° LSB. What does the resulting weight vector do?

phases = $\boxed{0, 0, 0, 0^\circ}$

Every one of -11.9°, $+3.3^\circ$, -3.3° and $+11.9^\circ$ is within 45° of zero, so all four round to 0°.

$$0^\circ, \quad 0^\circ, \quad 0^\circ, \quad 0^\circ$$

The phase information is gone entirely and the array is left as the amplitude taper of part 1(c), with no null at all.

   (d) Explain why holding a deep null is more demanding of phase resolution than holding a beam peak.

At the beam peak all element contributions add in phase, and a small phase error reduces each contribution by cos of that error, which is a second-order loss. At a null the contributions are arranged to cancel exactly, so the residual is a difference of nearly equal quantities and any phase error appears in it directly, at first order. A 2.8° error costs roughly 0.01 dB at the peak and sets a floor near −48 dB in the null, which is deeper than the lab sweep can display but a floor all the same.

4. **Two-channel MVDR by hand.** The PHASER's two digital channels are steered so that a target off boresight presents the real steering vector $s = [1, -1]^T$, while an interferer on boresight presents $u = [1, 1]^T$ with power $P = 9$ relative to a per-channel noise power of 1. The covariance seen by the beamformer is $R = \sigma^2 I + P\,uu^T$.

   (a) Build R.

   $$uu^T = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 9\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 9 \\ 9 & 10 \end{bmatrix}$$

   $R = \boxed{\begin{bmatrix} 10 & 9 \\ 9 & 10 \end{bmatrix}}$

   (b) Compute the MVDR weights $w = R^{-1}s/(s^T R^{-1} s)$.

   The determinant is $100 - 81 = 19$, so

   $$R^{-1} = \frac{1}{19}\begin{bmatrix} 10 & -9 \\ -9 & 10 \end{bmatrix}$$
   $$R^{-1}s = \frac{1}{19}\begin{bmatrix} 19 \\ -19 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

   $$s^T R^{-1} s = (1)(1) + (-1)(-1) = 2$$
   $$w = \frac{1}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$$

   $w = \boxed{(0.5,\ -0.5)^T}$

   (c) Evaluate the beamformer's response to the target, $w^T s$, and to the interferer, $w^T u$.

   $w^T s,\ w^T u = \boxed{1,\ 0}$

   $$w^T s = (0.5)(1) + (-0.5)(-1) = 1$$
   $$w^T u = (0.5)(1) + (-0.5)(1) = 0$$

   Unit gain on the target, as the constraint requires, and a complete null on the interferer.

   (d) Which lab procedure did MVDR just reproduce, and what would w become if the interferer were switched off?

   The weights $[0.5, -0.5]^T$ subtract one channel from the other, which is procedure B: Beam 1 Phase set to 180°, the monopulse difference beam. MVDR was not told about boresight interference; it found the difference beam because that is the quietest combination that still holds unit gain on the target. With the interferer switched off, $R = I$ and $w = s/(s^T s) = [0.5, -0.5]^T$, the ordinary matched beamformer for this look direction, which here happens to be the same vector.

5. **Choosing an approach.** Each situation below uses the same 8-element PHASER.

(a) A fixed ground emitter sits at a surveyed bearing of $+35°$ from the array face and does not move. The array must keep its beam on a target at $-10°$. Which approach would you use, and why?

Manual null steering. The interferer's angle is known and fixed, so the weights can be computed once and left in place, and the calculation gets all eight analog elements to work with rather than the two digital channels. It also costs no digital processing and no snapshot collection. The main lobe pays a fraction of a decibel because the null is far from the beam.

(b) An airborne jammer of unknown and changing bearing appears during a mission. Which approach would you use, and why?

MVDR. Manual null steering needs the interferer's angle before it can compute anything, and by the time an operator measures the bearing and re-enters eight weights the jammer has moved. MVDR estimates the covariance from the received data and re-solves every few snapshots, so the null follows the jammer without anyone knowing where it is.

(c) A requirement calls for a 33 dB notch on a board whose coarse 3-bit phase shifters hold a null only to about 21 dB (Lesson 27). Taking roughly 6 dB of improvement per added bit, how many phase bits are needed?

$B = \boxed{\text{5 bits}}$

$$33\text{ dB} - 21\text{ dB} = 12\text{ dB} \quad \Rightarrow \quad \frac{12}{6} = 2 \text{ additional bits}$$

$$B = 3 + 2 = 5 \text{ bits}, \qquad \text{LSB} = \frac{360°}{32} = 11.25°$$

(d) Two jammers now illuminate the array from different bearings. State what happens when MVDR is asked to handle both on this hardware.

It cannot. Two digital channels give two degrees of freedom, one of which is spent on the look-direction constraint, leaving exactly one null. Facing two interferers the beamformer places its single null where it reduces total output power the most, which is generally a compromise between the two and a deep null on neither. Handling both would require a third digital channel, so the limit is the hybrid architecture rather than the algorithm.

Documentation: ______________________________